

Stress-Testing SynthID:

A Hands-On Comparison of GAN-Based and Regeneration-Based Attacks on AI-Generated Image Watermarks

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Abstract

SynthID [Google DeepMind, 2023] is Google DeepMind’s invisible image watermarking scheme deployed on the outputs of Gemini’s image generators (Nano-Banana / Gemini 2.5 Flash Image). It is intended to be robust under common image manipulations while remaining imperceptible. We probe its robustness via two attack families that bracket the realistic adversary model: (i) a *surrogate-classifier* adversarial GAN that imitates a SynthID-like binary detector and is trained to fool it with bounded perturbations, and (ii) the *diffusive-regeneration* attack of Zhao et al. [Zhao et al., 2023] as adapted by the WAVES benchmark [An et al., 2024], which round-trips an image through a public Stable Diffusion model at varying noising depths. The two attacks span the classical adversarial-ML axis (gradient-driven but reliant on a surrogate) and the distortion-based axis (gradient-free, decoder-agnostic). We replicate both attacks on a held-out set of 104 Nano-Banana-generated, SynthID-watermarked images and report attack success against an independently trained classifier that approximates SynthID’s behaviour, as well as against the deployed SynthID detector via Google AI Studio. We find that the surrogate-GAN attack achieves 99% classifier label-flip with imperceptible L_∞ -bounded perturbations but *does not* disrupt the actual SynthID watermark, while regeneration at moderate depth ($N=20 \dots 80$ DDIM steps) reliably erodes both the surrogate signal and the deployed detector’s output at a visible quality cost. The full code, the test set, the pre-computed attack outputs, and the prompts used to generate every test image are released as a hands-on kit at <https://github.com/rabbanitw/WAVES/tree/synthid-regen-kit>.

1 Introduction

Generative image models from Google, OpenAI, and others now ship with *invisible watermarks* that are embedded at generation time and intended to survive routine image manipulation. The most-deployed example is Google DeepMind’s SynthID [Google DeepMind, 2023], attached to every image produced by the Gemini image-generation family (Nano-Banana / Gemini 2.5 Flash Image; we use these terms interchangeably for the generator that produced our test set). SynthID is presented as imperceptible-but-detectable even after the kinds of post-processing a typical user applies: cropping, mild compression, colour correction. The detector itself is not public; it is exposed only through Google’s AI Studio.

This work targets the natural follow-up question: *how brittle is SynthID against attackers who do not have decoder access?* We frame two attack families that, between them, cover the realistic threat surface:

(1) Surrogate-classifier attacks. The adversary trains its own binary classifier as a stand-in for the real SynthID detector — in our case, a ResNet-18 fine-tuned to distinguish natural photographs from Nano-Banana outputs — and then trains a generator G to find an L_∞ -bounded perturbation that fools the surrogate. This is the GAN-against-surrogate-classifier family explored in WAVES [An et al., 2024] under the name *AdvCls*, and it is the natural endpoint of the classical adversarial-ML pipeline applied to a watermarking problem. The question it answers is: does an adversarial perturbation that fools a SynthID-shaped classifier also disrupt the deployed SynthID detector? On every existing watermarking benchmark, the answer is “rarely” [An et al., 2024, Zhao et al., 2023], but this remains the most visually-conservative attack and is therefore worth empirically grounding on the deployed SynthID stack.

(2) Diffusive-regeneration attacks. The adversary feeds the watermarked image into a publicly available diffusion model (we use Stable Diffusion 1.4 [Rombach et al., 2022]), forward-noises the latent for N scheduler steps, and reverse-denoises back to pixel space. The diffusion model has no knowledge of SynthID’s embedding mechanism, but the round-trip non-trivially perturbs the high-frequency structure that any spread-spectrum watermark needs to ride on. Zhao et al. introduced this attack in 2023 [Zhao et al., 2023]; WAVES [An et al., 2024] adopted it as their primary benchmarked attack under the name *Regen-Diff* and reported it as the most damaging single-stage attack against three different academic watermarks (Tree-Ring, Stable-Signature, Stega-Stamp). The question this family answers is: does the same attack mechanism continue to work against the production SynthID stack?

Our contribution is a side-by-side, reproducible empirical comparison of both attack families on the deployed SynthID detector, against a common held-out target set of 104 Nano-Banana-watermarked images. Our findings line up with the existing literature (An et al. 2024, §F.3.2): the GAN attack defeats our SynthID-shaped surrogate at near-perfect rate (99.0% label-flip on independently-held classifier, mean LPIPS 0.012) but leaves the deployed SynthID detector essentially unmoved, while diffusive-regeneration steadily erodes the deployed detector’s output as N grows. We also release the kit (code, 104-image test set with prompts, pre-rendered regen outputs at $N = 10, 20, 40, 80$, and the worked-negative-example GAN training code) as a hands-on package for the DEFCON 34 AI Village.

Test set. We curated a held-out set of **104** SynthID-watermarked images (512×512 JPEG, generated by Nano-Banana via the Gemini 2.5 Flash Image API). The generation prompts (one per image, indexed 0–103) were sourced from a list of 150 short “camera+lens+subject” phrases of the form: “*jstyle prefixj*: Photo shot on camera: *jsubject descriptionj*” (e.g., “*Disposable camera flash photo, 1990s style: Photo shot on camera: side view of a brown horse wearing a saddle. The number 55 is stamped in white on its rear flank.*”). The style-prefix loadout (“Nikon film grain”, “Kodak Portra”, “Petzval lens”, “Hyperfocal landscape”, etc.) is necessary to coerce Nano-Banana into photoreal output — without it, many prompts default to illustrative or cartoon styles. The 104-image test set, the full 150-line prompt list, and per-image regen outputs at $N=10, 20, 40, 80$ are all shipped in the released kit.

2 Surrogate-Classifer GAN Attack

The surrogate-classifier attack splits into two independently-trained components: a binary classifier C_{eval} that mimics SynthID’s public-facing behaviour (“is this image AI-generated?”) and an

adversarial generator G that fools C_{eval} within a bounded perturbation budget. This factoring is the WAVES [An et al., 2024] AdvCls recipe.

2.1 Surrogate classifier C_{eval}

We use Apple’s Pico-Banana-400K dataset [Qian et al., 2025] as training data for C_{eval} . Pico-Banana-400K supplies $\sim 258\text{K}$ paired (real Open Images photo, Nano-Banana-edit) triples from the SFT split, each tagged with one of 35 edit-type labels. We filter to the $\sim 174\text{K}$ *photoreal* edit types (excluding “cartoonify”, “LEGO-ify”, “Van-Gogh style”, and the $\sim 30\%$ other stylized categories) and downsample to a 3,447-pair working subset, partitioned as **train**=2,947, **val**=200, **test**=300. We also evaluate on the 104-image held-out Nano-Banana test set described in Section 1, which is from a different prompt distribution (text-to-image, not instruction-edit) and therefore is OOD relative to C_{eval} ’s training distribution.

Codec-matched preprocessing. The original-class images (Pico-Banana sources) are stored as JPEG; the NB-edited images are stored as PNG. A naive classifier rapidly learns to discriminate JPEG-artefact signatures from PNG cleanness and never sees the actual watermark or stylistic NB fingerprint. We therefore re-encode both classes through JPEG quality 95 in memory before feeding either to the classifier (and identically inside the GAN’s data pipeline). All numbers in this paper are with codec matching enabled.

Architecture and training. C_{eval} is a ResNet-18 [He et al., 2016] pretrained on ImageNet, with its 1000-way head replaced by a 2-way linear classifier (label 0 = real photo, label 1 = NB edit). Inputs are resized to 256×256 and normalised with the ImageNet mean/std. We train for 20 epochs at batch size 64 with AdamW (lr 1×10^{-4} , weight decay 1×10^{-4}), cosine-annealed across $20 \times \lceil 2947/64 \rceil$ steps. Training is single-precision on a single A100. The best checkpoint by validation accuracy achieved **88.5%** held-out test accuracy (531/600 correct, balanced between the classes), with 94.0% recall on NB edits at the operating point. On the held-out 104-image SynthID test set, this classifier flags **91.3%** as label 1 (mean confidence $p=0.914$, median $p=1.000$), confirming the signal it picked up generalises to the deployed Nano-Banana family.

2.2 Generator G and adversarial training

We explore two surrogate-classifier attack variants. The first, included in the released kit, is a standard pix2pix-style GAN; the second is a stronger AdvGAN-style [Xiao et al., 2018] attack we ran to confirm the failure mode is not specific to the weaker attack.

Variant A: pix2pix GAN (released). The generator is a residual U-Net taken from the pix2pix implementation [Isola et al., 2017] (8 down-sampling blocks, 8 up-sampling blocks, $\sim 54\text{M}$ parameters), trained jointly against a 70×70 PatchGAN discriminator. Loss:

$$\mathcal{L}_G = \mathcal{L}_G^{\text{adv}} + \lambda_{\text{preserve}} \cdot \text{LPIPS}(G(x_{\text{edit}}), x_{\text{edit}}), \quad (1)$$

with LPIPS [Zhang et al., 2018] computed against the pretrained VGG backbone, $\lambda_{\text{preserve}} = 10$, 40 epochs, batch 16, Adam [Kingma and Ba, 2015] (lr 2×10^{-4} , $\beta_1=0.5$, $\beta_2=0.999$) on both generator and discriminator. Training is single-precision on a single A100, ~ 95 s/epoch (JPEG-re-encode-bound).

Variant B: AdvGAN-style with bounded perturbator. A flat fully-convolutional perturbator G (6 layers, no down-sampling, $\sim 0.6\text{M}$ parameters) outputs $\hat{x} = \text{clamp}(x + \epsilon \cdot \tanh(G(x)), -1, 1)$ with $\epsilon = 16/255$, enforcing an L_∞ -bounded perturbation. G is trained against an *independently-trained, frozen* surrogate classifier C_{train} (same recipe and seed offset as C_{eval} , 87.3% held-out test accuracy) with

$$\mathcal{L}_G = \text{CE}(C_{\text{train}}(G(x)), \text{label}=0) + \lambda_{\text{preserve}} \cdot \text{LPIPS}(G(x), x), \quad (2)$$

$\lambda_{\text{preserve}} = 1$, Adam (lr 2×10^{-4}), 20 epochs. This factoring — a held-out C_{train} used as the attack target plus a separate held-out C_{eval} used to score transfer — is the standard recipe for measuring *transferable* adversarial examples [Xiao et al., 2018, An et al., 2024]; if instead G trains directly against C_{eval} the resulting label-flip rate is training accuracy and uninformative.

Common training data. Both variants use the same 2,947-pair Pico-Banana-400K train split as C_{eval} . The respective held-out classifier used at evaluation time (C_{eval} for variant A; C_{eval} for variant B’s transfer score) is *never* touched during generator training, so reported label-flip rates measure transfer, not in-sample fitting.

Software stack. PyTorch 2.1.0 + CUDA 11.8, torchvision 0.16.0, diffusers 0.21.4, transformers 4.35.0, lpips 0.1.4, numpy 1.26.4, on a single A100-40GB.

2.3 Results

Table 1: GAN attack outcomes. “Label-flip” is the fraction of $G(x_{\text{edit}})$ predicted as label 0 (“real photo”) by the held-out C_{eval} on the 300-pair Pico-Banana test split. LPIPS is computed between $G(x_{\text{edit}})$ and x_{edit} with the same VGG backbone the training loss used. The rightmost column reports whether the perturbed image bypasses Google’s deployed SynthID detector (qualitative spot-check via AI Studio on a representative ~ 20 -image subset).

attack	label-flip on C_{eval}	LPIPS G -vs-input	flips deployed SynthID?
no attack (baseline edit-detection error)	6.0%	0.000	—
Variant A: pix2pix GAN ($\lambda=10$)	11.3%	0.031	no
Variant B: AdvGAN, $L_\infty=16/255$	99.0%	0.012	no

The qualitative difference between the two variants is large: Variant A barely fools C_{eval} (11.3% flip), and its perturbations take the form of locally-supported visible artifacts (“black blobs”) in the generated outputs — a known failure mode of pix2pix-style generators applied to bounded-perturbation tasks. Variant B reaches near-saturating 99% classifier-flip with imperceptible $L_\infty \leq 16/255$ perturbations.

The headline finding is that *neither variant disrupts the deployed SynthID detector*. A manual spot-check of perturbed images from both variants through Google’s AI Studio returns “SynthID watermark detected” at the same rate as un-attacked input. This empirically realises the WAVES paper’s observation [An et al., 2024, §F.3.2] that surrogate-classifier attacks are largely decoupled from the actual watermark embedding: G has learned the classifier’s idiosyncratic decision boundary, not SynthID’s payload, and the resulting perturbation lives in the wrong feature subspace to

disturb the watermark carrier. We include Variant A in the released kit as a worked negative example because the experiment *is* the lesson, with ~ 2 hours of A100 wall-clock from clone to repro; Variant B is documented here for completeness but is not in the kit.

3 Diffusive Regeneration Attack

We follow the symmetric-DDIM regen recipe of Zhao et al. [Zhao et al., 2023], as used in WAVES Section F.2: encode the image into the VAE latent space of a Stable Diffusion model, add scheduler noise at training timestep $N-1$, and reverse-denoise through the bottom N steps of a 1000-step DDIM schedule back to pixel space. Concretely we use the SD-1.4 checkpoint [Rombach et al., 2022] at fp16, the diffusers DDIM scheduler, classifier-free guidance scale 7.5, and an empty text prompt. Because each inference step of our schedule covers exactly one training timestep, the attack has a single free parameter N that controls both how much noise is added and how much budget the denoiser has to remove it. This is in contrast to WAVES’s published asymmetric variant (50-step inference schedule, $\sim N/20$ denoising iterations), which leaves residual noise in the output.

Outputs. Figure 1 shows the original $\rightarrow N=10 \rightarrow 20 \rightarrow 40 \rightarrow 80$ progression for three representative test images. At $N=10$ the watermarked image is visually unchanged; by $N=80$ fine structure is lost and visible softening / hallucinated detail starts appearing. Pre-rendered copies of all $104 \times 4 = 416$ attacked images are shipped with the kit at `images/regen_{10,20,40,80}/`.

Detector response. We score every attacked image with the held-out C_{eval} described in Section 2 (which generalises across NB pipelines, hitting 91.3% flag rate on the unattacked Nano-Banana test set). Detection collapses monotonically with N :

N (sym DDIM regen steps)	C_{eval} NB-flag rate	Δ from baseline
0 (no attack)	91.3% (95/104)	—
10	64.4% (67/104)	−26.9 pts
20	58.7% (61/104)	−32.6 pts
40	53.8% (56/104)	−37.5 pts
80	45.2% (47/104)	−46.1 pts

The trend matches the WAVES-reported curves for SD-1.x and Tree-Ring, Stable-Signature, and Stega-Stamp at comparable attack depths [An et al., 2024, §5.3]. Spot-checks against Google’s deployed SynthID detector confirm the directional agreement: at $N \geq 20$ a representative sample of attacked images come back as “no watermark detected”. The attack is thus the most reliable *decoder-agnostic* channel for SynthID removal we are aware of, and its main cost is the image-quality degradation visible in Figure 1.

4 Discussion

Surrogate-classifier attacks remain a dead end without decoder access. The 99% label-flip rate against C_{eval} is a powerful red herring. C_{eval} is high-accuracy (88.5% in-distribution, 91.3% on the held-out Nano-Banana set) and was independently trained from the GAN; the attack’s failure to transfer to the deployed SynthID detector is therefore not an artifact of a weak surrogate or training leakage. It reflects the WAVES finding [An et al., 2024] that SynthID’s embedding scheme



Figure 1: Symmetric-DDIM diffusive regeneration of three test images at $N \in \{10, 20, 40, 80\}$ (left-to-right; the leftmost panel is the unattacked original). SD-1.4, fp16, classifier-free guidance 7.5, empty prompt, seed 1024. By $N=80$ the regen has both removed the SynthID watermark (verified manually via Google AI Studio) and visibly degraded the image’s high-frequency structure.

operates on a representation that is not recoverable by an ImageNet-pretrained binary classifier of the “natural photo vs. AI-edit” kind, and that gradient-based attacks on this surrogate live in a different feature subspace than SynthID itself.

Regeneration trades quality for watermark removal, but it works. At $N=80$ the deployed detector is fooled with high probability, but the image’s high-frequency detail is also visibly degraded (Figure 1, rightmost column). This is the classical robustness/quality tension: SynthID, like any spread-spectrum watermark, can be removed by any image processing that disturbs its carrier band, but at a quality cost the attacker may not want to pay. The remaining question for future work is whether the GAN approach can be re-cast to *repair* the regen-induced damage — i.e., stack a quality-restoring generator on top of a low- N regen — to give the attacker a low-distortion, decoder-agnostic SynthID remover. Existing preliminary attempts in this direction (“RinseD-VAE”, “RegenD-KLVAE” in WAVES §F.2.2) suggest the trade-off is achievable but not yet on a deployed-watermark scale.

Reproducibility. The full release kit — <https://github.com/rabbanitw/WAVES/tree/synthid-regen-kit> — contains the 104-image test set, the 104 generation prompts, the pre-rendered 416 attacked images, the GAN training and evaluation code, and the regeneration CLI. All numbers above are pinned to deterministic seeds (PyTorch / numpy seed 42 for the GAN, seed 1024 for regen) and

re-run identically on an A100. The pipeline depends on diffusers 0.21.4 and the specifications of the older `StableDiffusionPipeline` API; newer diffusers will require re-wiring the vendored `ReSDPipeline` subclass.

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